

Joshua Watkins

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SUMMARY

I enjoy working with data because it allows me to turn complex information into something meaningful for the business. As a Senior Data Engineer, I focus on building scalable data pipelines and machine learning systems that help teams move from raw data to clear insights and predictions they can actually use.

In my recent work, I built end-to-end data and ML pipelines that reduced data processing time by about 40% and improved forecasting accuracy by roughly 20–25%. These improvements helped the company make faster operational decisions and contributed to several million dollars in cost savings through better resource planning and operational efficiency.

What motivates me most is seeing the real impact of good data engineering—when reliable data and ML models help teams understand their customers better, optimize operations, and stay ahead in a competitive market. Ready to achieve big things together!

SKILLS

Languages: Python, SQL, PySpark, Spark SQL

Data Engineering: ELT pipelines, batch processing, streaming pipelines, large scale data pipeline development, workflow orchestration, Apache Airflow

Databricks Platform: Databricks Lakehouse, Delta Lake, Delta Live Tables, Databricks SQL, Unity Catalog, MLflow

Data Lake & Storage: Data Lake architecture, Delta Lake, Parquet, ORC, Avro, Amazon S3

Data Warehousing: Amazon Redshift, Snowflake, Databricks SQL Warehouse, dimensional modeling, star schema, query optimization

Machine Learning & Analytics: ML pipelines, feature engineering, model training pipelines, MLflow experiment tracking, predictive analytics

Big Data Processing: Apache Spark, distributed computing, large scale data processing

Cloud Platforms: Azure Data Lake Storage, Azure Databricks, Azure Data Factory, Azure Synapse Analytics, Azure Functions, Azure Event Hubs, Azure Key Vault, Azure Monitor, AWS S3, AWS Glue, Google Cloud Storage, BigQuery

DevOps & Platform: CI/CD pipelines, Git, Docker, Kubernetes, Terraform

Data Governance: Data lineage, metadata management, data quality validation, access control, encryption at rest, encryption in transit

PROFESSIONAL EXPERIENCE

Accenture Houston | Senior ML Data/Analytics Engineer

Houston, TX | Jan 2022 – current

- Led development of a Cloud Customer Intelligence Platform, integrating multi-source customer interaction, transaction, and behavioral data into an Azure Databricks Lakehouse architecture to support advanced analytics and machine learning, enabling the client to improve customer insights and revenue growth strategies
- Designed scalable data ingestion pipelines using Azure Data Lake Storage (ADLS), Azure Data Factory, and Databricks Auto Loader to capture structured and semi-structured data from CRM systems, APIs, and transaction databases, processing several terabytes of daily data and improving data availability for analytics teams by over 35%

- Built distributed transformation pipelines using Azure Databricks, PySpark, and Delta Lake to cleanse, standardize, and aggregate large customer datasets, delivering curated analytics tables and feature datasets that supported ML models and business intelligence dashboards
- Developed machine learning data pipelines and feature engineering workflows using Databricks MLflow and Azure-based data services, improving customer segmentation and predictive analytics accuracy by about 22%, enabling marketing teams to optimize targeting strategies
- Implemented automated orchestration and monitoring using Apache Airflow and Azure Monitor, improving pipeline reliability and reducing data processing time by roughly 40%, which helped the client reduce operational costs by several million dollars annually
- Optimized analytics delivery using Databricks SQL and Azure Synapse Analytics integrations to provide near real-time reporting and customer insights to business teams, contributing to measurable improvements in campaign performance and generating positive feedback from client stakeholders and end users

Wells Fargo | Cloud/Big Data Engineer

Charlotte, NC | Jul 2017 - Dec 2021

- Built a Cloud Risk & Transaction Analytics Platform to centralize banking transaction, customer, and fraud monitoring data using an Azure Databricks Lakehouse architecture, enabling faster fraud detection and improving data accessibility for risk and compliance teams
- Designed scalable data ingestion pipelines using Azure Data Lake Storage (ADLS), Azure Data Factory, and Databricks Auto Loader to capture high-volume transaction and customer activity data from core banking systems and APIs, processing multi-terabyte daily datasets and improving data availability for analytics teams by over 30%
- Developed large-scale data processing pipelines using Azure Databricks, PySpark, and Delta Lake to cleanse and standardize transaction datasets, creating curated analytics tables that supported fraud analytics, risk monitoring, and regulatory reporting
- Implemented batch and near real-time processing using Databricks Structured Streaming and Azure Event Hubs, enabling faster detection of suspicious activity and helping fraud teams reduce investigation response time by approximately 35%
- Automated pipeline orchestration and monitoring using Apache Airflow, Azure Monitor, and CI/CD pipelines, improving pipeline reliability and reducing data processing latency by about 40%, contributing to millions of dollars in prevented fraud losses
- Optimized analytics delivery using Databricks SQL and Azure Synapse Analytics integrations with enterprise reporting tools, providing risk analysts and compliance teams with trusted datasets and dashboards, improving decision-making speed and receiving strong positive feedback from internal business stakeholders and fraud operations teams

Jack Henry & Associates | Data Engineer / ETL Developer

Plano, TX | May 2015 - Jun 2017

- Built a Banking Data Integration Platform to consolidate core banking, payment, and customer data into an Azure-based data lake using Azure Data Lake Storage and Azure Databricks for large-scale processing, enabling faster reporting and improving data accessibility for financial analytics teams
- Designed and implemented ELT data ingestion pipelines using Azure Data Lake Storage (ADLS), Azure Data Factory, and Databricks to collect transaction and account data from core banking databases and batch file feeds, processing high-volume financial datasets and improving data availability by over 30%
- Developed scalable transformation pipelines using Azure Databricks, PySpark, and Delta Lake to cleanse, standardize, and structure banking data into analytics-ready datasets, supporting operational reporting and customer financial insights
- Implemented automated workflow orchestration using Apache Airflow and Azure services to schedule and monitor ETL jobs, improving pipeline reliability and reducing data processing time by approximately 35%
- Optimized storage and query performance using Databricks SQL and Azure Synapse Analytics to enable faster reporting for internal banking teams and improve decision-making speed across operations

- Delivered trusted datasets for analytics and regulatory reporting, helping business teams identify operational inefficiencies and improve service response time, which contributed to cost savings and positive feedback from internal banking clients and stakeholders

EDUCATION

Baylor University - Waco, TX | B.S.Computer Science | Aug 2011 – May 2015

CERTIFICATIONS

Microsoft Certified Azure Solutions Architect Expert
Microsoft Certified Azure Data Engineer Associate (DP-203)
Microsoft Certified Fabric Analytics Engineer Associate (DP-600)
Databricks Data Engineer Professional
Databricks Lakehouse Platform Architect
Snowflake SnowPro Advanced Data Engineer
Certified Kubernetes Administrator CKA